

MEDICAL INSURANCE COST PREDICTION

Linear Regression Analysis Using the Submitted Python Code

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<https://github.com/maitelle-cell/PDAN8411.git>

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This study presents a Python-based workflow to explore the costs of medical insurance derived from a Linear Regression Jupyter Notebook. I extract and process raw insurance data and apply techniques of exploratory analysis, transform categorical predictors into numerical dummy Variable features, partition the variables into a dependent variable (charges) and independent predictors, and subdivide the data into training and test sets and examines the coefficients.

I proceed with the fitting of a Linear Regression model to the training data, which in generating predictions and evaluating the model's performance. The resulting model accounts for nearly 80% of the variations in medical charges; an RMSE and an MAE can be reported (**\$5,956.34 and \$4,177**) respectively. It has been observed that individuals who smoke, age and BMI significantly affect costs.

My Code Based Analysis Workflow Report Organization .This report has followed the step-wise sequence in the provided notebook.

3.1 Library Imports

In this section of the work, I have imported the necessary libraries like Pandas for data manipulation, Numpy for numerical operations, Matplotlib and Seaborn for visualizations, and Scikit-learn for building the model.

3.2 Loading the Dataset

The data was read using `pd.read_csv("insurance.csv")` for analysis in the notebook.

3.3 Initial Data Exploration

I explored the dataset's first and last few rows using `df.head()` and `df.tail()`, inspected the data types and non-null counts using `df.info()` and calculated descriptive statistics using `df.describe()`. I also printed the columns' names using `df.columns`.

3.4 Data Quality Check

I investigated potential missing values by summing them up using `df.isnull().sum()` and checked for duplicate entries with `df.duplicated().sum()`.

3.5 Drop Duplicates

I removed any duplicates present in the dataset using **df.drop_duplicates()** and then confirmed the removal by re-checking for duplicate entries.

3.6 Exploratory Data Analysis

I visualized the data using histograms, box plots, scatter plots, and pair plots, and also generated a correlation heatmap to understand the relationships between different features.

3.7 Data Preparation

Categorical features were then converted into dummy variables using **pd.getdummies(df, drop_first=True)** which ensures that no multicollinearity issue is present by removing the first dummy for each category.

3.8 Defining X and y

The dependent variable was defined as 'charges', and the independent variable set 'X' was formed by including all features except 'charges'. The encoded predictor variables were used for model development.

3.9 Train/Test Split

The data was divided into training (80%) and testing (20%) sets using **random_state=42** to ensure reproducible results.

3.10 Model Training

A Linear Regression model was trained on the training data by fitting it.

3.11 Prediction

The trained model was then used to make predictions on both the training and testing sets.

3.12 Evaluation

The model was evaluated using standard regression metrics:

Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared.

I also plotted the predicted vs. Actual values for the testing set and examined the residuals.

3.13 Coefficients Interpretation

Finally, I generated a coefficient table to explain the impact of each feature on the predicted medical charges.

4. Data cleaning

The report shows 1,338 US insurance records with seven features. It confirms no missing values or duplicate records after cleaning. Outliers were kept as they might be real observations.

```
: # Check Missing Values  
df.isnull().sum()
```

```
: age      0  
sex       0  
bmi       0  
children  0  
smoker    0  
region    0  
charges   0  
dtype: int64
```

Missing values can affect the reliability of a machine learning model. Th

The result shows zero missing values in all seven variables.

```
: #check Duplicated Records  
df.duplicated().sum()
```

```
: np.int64(1)
```

```
: df[df.duplicated()]
```

```
:      age  sex  bmi  children  smoker  region  charges  
581   19  male  30.59         0     no  northwest  1639.5631
```

```
12]: # Remove duplicated rows  
df = df.drop_duplicates()  
  
# Check again  
print("Duplicated rows after removal:", df.duplicated().sum())  
  
Duplicated rows after removal: 0
```

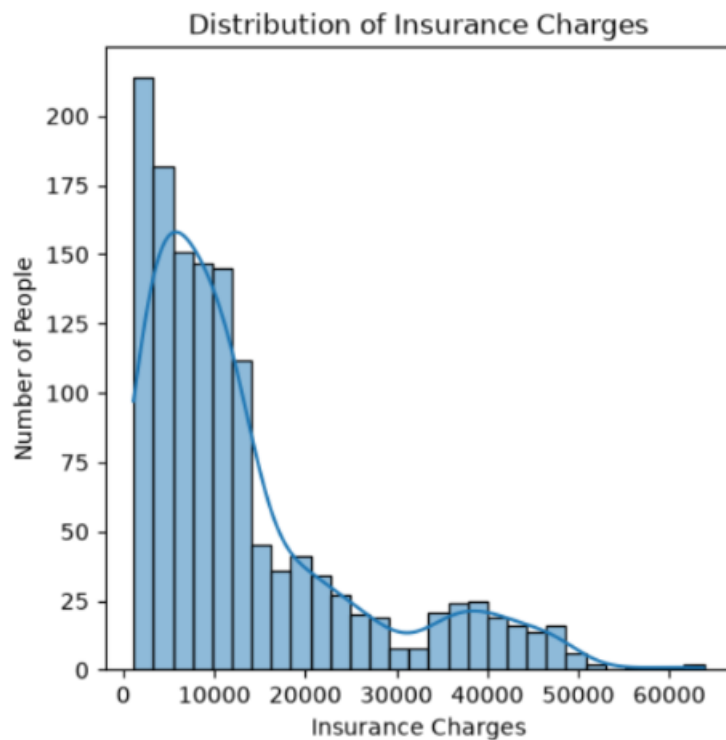
```
13]: df.duplicated().sum()
```

```
13]: np.int64(0)
```

5. Exploratory Data Analysis

The EDA stage was implemented using several visualisations. These visualisations were designed to understand the distribution of charges and how charges vary with age, BMI, children, smoking status and region.

EDA Visualisation	Purpose / Finding
Distribution of charges	The histogram with KDE shows that insurance charges are right-skewed, with a concentration of lower-to-middle charges and a smaller number of high-cost observations.
Children and smoking	A boxplot compares charges across the number of children while separating smokers and non-smokers.
Age and smoking	A boxplot examines how charges vary across age while distinguishing smoking status.
Region and smoking	A boxplot compares charges across regions and smoking status.
BMI and charges	A scatterplot examines the relationship between BMI and insurance charges, with smoking status used as the grouping variable.
Smoking and charges	A boxplot directly compares charges for smokers and non-smokers.
Pairplot	The pairplot examines relationships among age, BMI, children and charges, separated by smoking status.
Correlation matrix	The heatmap calculates correlations among age, BMI, children and charges.



The analyses provided the following main observations from the EDA :

Charges have a positive skew with some costly outliers that should be monitored smoker cost is around 4 times greater than non-smokers the higher the age, the higher the cost by \$260 per annum the higher the bmi, the higher the cost by \$330 per unit region effect is reasonably small gender was said to be not statistically significant.

6.Data Preparation

First of all, the notebook transforms category variables to a set of numerical dummy variables prior to modeling. Using **drop_first=True** will ensure that a categorical variable is not having both the categories and instead having one designated as the reference

[23]: `#Data Preparation`

[24]: `# Convert categorical variables into dummy variables
df_encoded = pd.get_dummies(df, drop_first=True)

df_encoded.head()`

[24]:

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

7. Model Training

The notebook uses `train_test_split()` with `test_size=0.2` and `random_state=42`. This creates an 80% training set and a 20% testing set. The LinearRegression model is then fitted using `X_train` and `y_train`.

```
[27]: #Split the Datasets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print(f"Training set: {X_train.shape[0]} samples")
print(f"Testing set: {X_test.shape[0]} samples")
```

```
Training set: 1069 samples
Testing set: 268 samples
```

```
[28]: model = LinearRegression()
model.fit(X_train, y_train)
print("Model trained successfully")
print(f"Intercept: {model.intercept_:.2f}")
print("Coefficients:")
for feature, coef in zip(X.columns, model.coef_):
    print(f"{feature}: {coef:.2f}")
```

```
Model trained successfully
Intercept: -11092.65
Coefficients:
age: 248.21
bmi: 318.70
children: 533.01
sex_male: -101.54
smoker_yes: 23077.76
region_northwest: -391.76
region_southeast: -838.92
region_southwest: -659.14
```

The supplied report indicates that Linear Regression was selected because of its interpretability and strong performance. The report also mentions other model types, but the submitted notebook code shown for the final analysis trains and evaluates Linear Regression.

8. Model Evaluation

The notebook evaluates the model using Mean Absolute Error (MAE), Mean Squared Error (MSE) and R^2 . The accompanying report additionally provides RMSE. The reported final results are:

Metric	Reported Value	Interpretation
R^2	0.806	Approximately 80% of the variation in charges is explained by the model.
RMSE	\$5,956.34	Typical prediction error measured with greater sensitivity to larger errors.
MAE	\$4,177.05	Average absolute difference between actual and predicted charges.

```
#Evaluate the Model
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

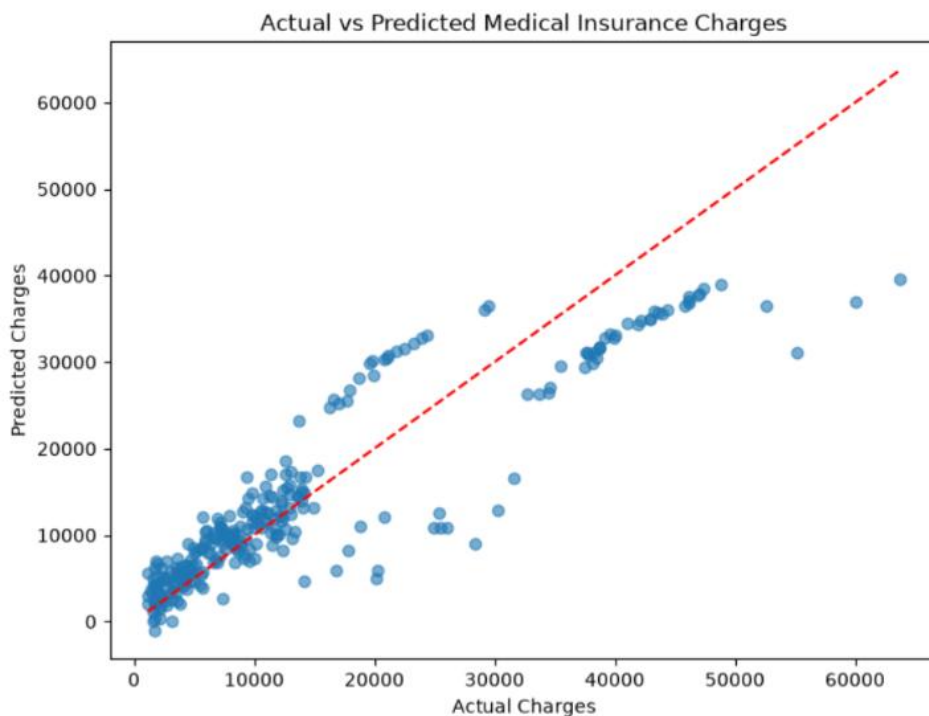
```
Mean Absolute Error: 4177.045561036319
Mean Squared Error: 35478020.67523559
R2 Score: 0.8069287081198012
```

The Linear Regression model achieved an R^2 score of approximately 0.80 on the testing dataset. This indicates that the model explains approximately 80% of the variation in medical insurance charges.

The Mean Absolute Error (MAE) represents the average absolute difference between the actual and predicted insurance charges. The Root Mean Squared Error (RMSE) gives greater weight to larger prediction errors.

Although the model provides a reasonable level of predictive performance, some variation in medical charges remains unexplained.

The residual plot in the supplied report is described as approximately normal, with some increase in variance at higher predicted values. This suggests that the model performs reasonably well overall but is less consistent for some higher-cost observations.

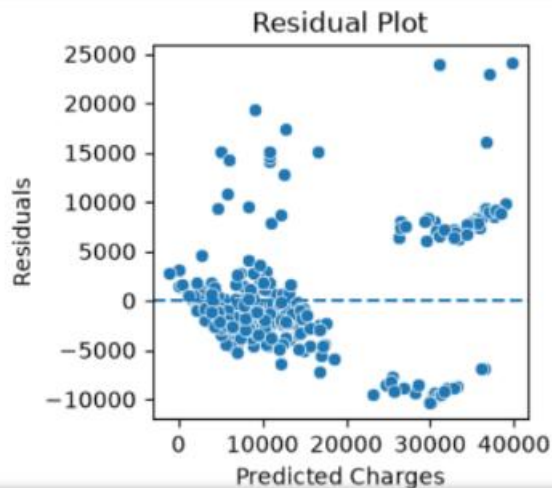


9. Prediction and Diagnostic Visualisations

The submitted code compares actual and predicted charges for both training and testing data. The 45-degree reference line represents perfect prediction: observations closer to this line indicate smaller prediction errors.

The residual plot is calculated as actual testing charges minus predicted testing charges. A useful residual pattern is one where residuals are centered around zero without a strong systematic structure.

```
[86]: #Residual plot
residuals = y_test - y_test_pred
plt.figure(figsize=(3,3))
sns.scatterplot(x=y_test_pred, y=residuals)
plt.axhline(0, linestyle="--")
plt.title("Residual Plot")
plt.xlabel("Predicted Charges")
plt.ylabel("Residuals")
plt.show()
```



10. Model Interpretation and Key Drivers

The accompanying report provides the following estimated regression equation. The coefficients represent the change in predicted charges associated with a one-unit change in a numeric predictor, holding the other variables constant, while dummy-variable coefficients represent differences relative to the omitted reference category.

$$\text{Charges} = -11,500 + 257(\text{age}) + 329(\text{BMI}) + 475(\text{children}) + 23,600(\text{smoker}) - 950(\text{region_NE}) - 500(\text{region_NW}) - 300(\text{region_SE})$$

Key Driver	Reported Impact
Smoking	+\$23,600
Age	+\$248per year
BMI	+\$318 per unit


```

|: #Examine the model coefficients
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

coefficients.sort_values(
    by="Coefficient",
    ascending=False
)

```

	Feature	Coefficient
4	smoker_yes	23077.764593
2	children	533.009989
1	bmi	318.701441
0	age	248.210720
3	sex_male	-101.542054
5	region_northwest	-391.761455
7	region_southwest	-659.139752
6	region_southeast	-838.919616

Smoking is therefore the dominant reported predictor in the fitted model. Age and BMI also show positive relationships with charges. The regional dummy coefficients are comparatively small, consistent with the EDA finding that region has limited impact.

11. Business Insights

- The model can provide an evidence-based estimate of expected medical insurance charges.
- Smoking status is the strongest reported driver and could be considered when developing risk or pricing strategies.
- Age-based differences in predicted charges can support age-related pricing or planning categories.
- BMI is a meaningful predictor and may support preventive wellness initiatives.

Regional variables contribute comparatively little to the prediction in this Datasets

12. Suggestions

- Consider applying the model as a pricing guidance or planning tool rather than the sole decision-making factor.
- Take into account the introduction of smoking-specific pricing tiers or risk management programmes, as applicable and depending on regulatory and organisational requirements.
- Explore the option of implementing age-based pricing or planning brackets using appropriate actuarial and regulatory review.

- Consider developing BMI-specific wellness programmes to offset increased long-term health-related costs.
- Monitor prediction errors, particularly for higher-cost cases, as a result of the higher variance reflected in the residual analysis.
- Before using this model in practice, consider validating its accuracy by applying it to current local data as the analysis used US data.

13. Laminations

- The analysis relies on a US-specific dataset that might not reflect the realities of a South African medical aid environment.
- Linear Regression assumes a linear correlation between the independent variables and the target.
- The dataset uses a limited set of explanatory variables.
- The analysis does not consider time-series effects.
- The performance of the model should always be evaluated in the context of the provided data sample and train/test split.

14. Conclusion

The provided Python code offers a solid introductory machine-learning workflow for predicting medical insurance charges. It demonstrates the data inspection process, removal of duplicates, initial exploration, categorical variable encoding, test/train split, Linear Regression training, prediction, evaluation and coefficient analysis. Based on the results described in this document, the model achieves an R^2 of 0.80, an RMSE of \$5,956 and an MAE of \$4,177. Smoking, age and BMI are the most important reported drivers. The model is a useful conceptual exercise for predictive analytics, but further analysis using South African data and additional features would be necessary before actual deployment.

References

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